Agents and Environments in AI
AI System Composition
Agent
Perceives environment
Acts upon environment
Sensors
Effectors/Actuators
Environment
Surrounds agent
Not part of agent
Situation agent is present in
Where agent lives, operates
Provides something to sense and act upon
Properties of Environment
Discrete / Continuous
Discrete: Limited, distinct states (e.g., chess)
Continuous: Unlimited states (e.g., driving)
Observable / Partially Observable
Observable: Complete state from percepts
Partially Observable: Incomplete state
Static / Dynamic
Static: Does not change while agent acts
Dynamic: Changes while agent acts
Single agent / Multiple agents
May contain other agents (same or different kind)
Accessible / Inaccessible
Accessible: Sensory apparatus accesses complete state
Deterministic / Non-deterministic
Deterministic: Next state determined by current state + actions
Non-deterministic: Next state not completely determined
Episodic / Non-episodic
Episodic: Each episode independent, no future impact
Non-episodic: Actions in previous episodes affect subsequent ones
Categories of Agents
Human Agent
Sensors: Eyes, ears, nose, tongue, skin
Effectors: Hands, legs, mouth
Robotic Agent
Sensors: Cameras, infrared range finders
Effectors: Motors, actuators
Software Agent
Sensors: Encoded bit strings (programs)
Effectors: Actions (programs)
Agent Terminology
Performance Measure: Criteria for agent success
Behavior of Agent: Action after percept sequence
Percept: Agent's perceptual input at instance
Percept Sequence: History of all perceived inputs
Agent Function: Map from precept sequence to action
Rationality
Reasonable, sensible, good judgment
Concerned with expected actions and results
Performing actions for useful information
Ideal Rational Agent
Maximizes performance measure
Based on percept sequence
Based on built-in knowledge base
Depends on
Performance measures
Agent's percept sequence
Agent's prior knowledge
Actions agent can carry out
Right Action
Agent most successful in given percept sequence
Problem Characterized by PEAS
Performance Measure
Environment
Actuators
Sensors
Structure of Intelligent Agents
Agent = Architecture + Agent Program
Architecture: Machinery agent executes on
Agent Program: Implementation of agent function
Simple Reflex Agents
Actions based on current percept only
Rational if decision based on current percept
Environment completely observable
Condition-Action Rule: Maps state to action
Model Based Reflex Agents
Use world model to choose actions
Maintain internal state
Model: Knowledge of 'how things happen'
Internal State: Representation of unobserved aspects
Updating State Requires
How world evolves
How agent's actions affect world
Goal Based Agents
Choose actions to achieve goals
More flexible than reflex agents
Goal: Description of desirable situations
Utility Based Agents
Choose actions based on preference (utility) for states
Goals Inadequate When
Conflicting goals
Uncertainty of achievement
Nature of Environments
Artificial Environment
Confined to keyboard input, database, file systems
Turing Test environment
Rich, Unlimited Softbots Domains
Software agents (softbots)
Detailed, complex environment
Long array of actions in real time
Mixed Environment
Softbot scanning online preferences (real + artificial)